A SMART DUSTBIN WITH AUTOMATIC BIN INDICATION CAN HELP ENSURE CORRECT BIOMEDICAL WASTE SEGREGATION AND IMPROVE WASTE MANAGEMENT EFFICIENCY

A MINI – PROJECT REPORT

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ABSTRACT

Waste management is an important issue in modern cities and public places. Improper disposal of waste can lead to environmental pollution and health problems. To address this issue, a smart dustbin system is designed using Arduino Uno, IR Sensor Module, Inductive Proximity Sensor, and Servo Motor SG90. The main objective of this project is to develop an automatic and hygienic waste disposal system with basic waste segregation capability. The system detects the presence of waste using an IR sensor placed near the lid of the dustbin. When a person brings waste close to the bin, the IR sensor senses the object and sends a signal to the Arduino microcontroller. The Arduino processes this signal and activates the servo motor to open the lid automatically, allowing the waste to be disposed of without physical contact. After the waste enters the dustbin, the inductive proximity sensor checks whether the material is metallic. If metal is detected, the system directs the waste into the metal compartment; otherwise, it is directed to the general waste compartment. This automatic segregation helps in improving recycling efficiency and reducing manual sorting efforts. The servo motor plays a key role in controlling the lid and the internal segregation mechanism based on the sensor inputs. The system is simple, cost-effective, and easy to implement using basic electronic components. It can be installed in hospitals, laboratories, public places, and smart city environments to improve sanitation and waste management practices. Overall, the proposed smart dustbin enhances cleanliness, promotes automated waste segregation, and contributes to a cleaner and healthier environment.

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

CHAPTER 1

INTRODUCTION

Biomedical Waste (BMW) refers to the waste generated during the diagnosis, treatment, or immunization of humans and animals, as well as in research and healthcare activities. This type of waste includes infectious materials, sharps, chemicals, and other hazardous substances that pose serious risks to human health and the environment if not managed properly.

Effective biomedical waste management is essential for preventing the spread of infections and ensuring environmental safety. According to reports by the **World Health Organization**, nearly 75–90% of biomedical waste is non-hazardous, while the remaining 10–25% is hazardous and requires careful handling and disposal. In countries like India, the generation of biomedical waste is increasing rapidly due to the expansion of healthcare facilities and population growth, creating significant challenges in waste management systems.

Traditional methods of biomedical waste management involve manual segregation, collection, transportation, and disposal. However, these methods are often inefficient, time-consuming, and prone to human error, leading to improper handling and environmental contamination. The COVID-19 pandemic further highlighted these challenges, as there was a sudden surge in biomedical waste generation, putting additional pressure on existing waste management systems.

To overcome these limitations, modern technologies such as **Artificial Intelligence (AI)** and the **Internet of Things (IoT)** are being integrated into waste management systems. Smart bin systems equipped with sensors and automated mechanisms can detect, classify, and segregate waste efficiently with minimal human intervention. These technologies not only improve accuracy and safety but also enable real-time monitoring and data analysis.

This project focuses on the development of an AI and IoT-based smart biomedical waste management system that enhances waste segregation, reduces human exposure to hazardous materials, and promotes sustainable and efficient waste disposal practices.

1.1 EVOLUTION OF DUSTBIN USAGE IN WASTE MANAGEMENT

Waste disposal practices have evolved significantly over the years. In earlier times, dustbins were simple containers with minimal design and no technological support. The following explains how dustbins were used in the past and their limitations.

1.1.1 Open Waste Disposal System

In the early stages, people disposed of waste in open areas without using proper dustbins. Waste was thrown on streets, empty lands, or pits.

Explanation:

- No structured waste collection system
- No segregation of waste (all waste mixed together)
- Caused foul smell, pollution, and spread of diseases
- Attracted animals and insects

FIG 1.1 OPEN WASTE DISPOSAL SYSTEM

1.1.2. Basic Manual Dustbins

Later, simple dustbins were introduced in homes, hospitals, and public places.

Explanation:

- Made of metal or plastic
- Used only for storing waste temporarily
- Required manual handling for disposal
- No separation of waste types

FIG 1.2 BASIC MANUAL DUSTBINS

1.1.3. Community Dustbin System

In this system, large dustbins were placed in common public areas for shared use.

Explanation:

- Used by multiple households or facilities
- Waste collected periodically by municipal workers
- No proper segregation (mixed waste)
- Often overflowed due to irregular collection

FIG 1.3 COMMUNITY DUSTBIN SYSTEM

1.1.4. Early Segregation Practices

With increased awareness, basic waste segregation methods were introduced.

Explanation:

- Use of color-coded bins (e.g., for biodegradable and non-biodegradable waste)
- Limited awareness among users
- Segregation still done manually
- Errors were common due to lack of training

1.1.5 Manual Biomedical Waste Bins in Hospitals

Hospitals began using specific bins for biomedical waste.

Explanation:

- Color-coded bins for different waste categories (yellow, red, blue, etc.)

- Waste handled by healthcare workers manually
- High risk of infection and contamination
- No automation or monitoring system

FIG 1.4 MANUAL BIOMEDICAL WASTE BINS IN HOSPITALS

1.1.6 Ultrasonic-Based Waste Monitoring System

This system uses an ultrasonic sensor and an LCD display to monitor the fill level of biomedical waste bins in a simple and efficient way. The ultrasonic sensor is placed at the top of the bin to measure the distance between the sensor and the waste. As the bin fills, the distance decreases, allowing the system to determine how full the bin is.

A microcontroller (such as Arduino) processes this data and calculates the waste level. The information is then displayed on an LCD screen in the form of percentage or simple messages like "EMPTY," "HALF FULL," and "FULL." When the bin reaches its maximum capacity, the LCD displays "BIN FULL," indicating that it needs to be emptied.

This system reduces the need for manual inspection, prevents overflow, and helps maintain cleanliness in hospitals. It is a cost-effective and easy-to-implement solution for basic biomedical waste monitoring.

FIG 1.5 ULTRASONIC-BASED WASTE MONITORING SYSTEM

CHAPTER 2
LITERATURE SURVEY

Biomedical Waste (BMW) management is a critical aspect of healthcare systems due to its potential impact on human health and environmental safety. According to the World Health Organization (WHO), around 75–90% of biomedical waste is non-hazardous, while the remaining 10–25% is hazardous and requires special handling and disposal methods. Improper management can lead to infections, environmental pollution, and public health risks.

2.1 TRADITIONAL BIOMEDICAL WASTE MANAGEMENT

Conventional BMW management methods involve segregation, collection, transportation, treatment, and disposal. These processes are mostly manual and depend on staff compliance and awareness. However, these systems often suffer from human errors, lack of monitoring, and inefficient waste segregation. During the COVID-19 pandemic, the sudden increase in waste generation exposed the limitations of these traditional approaches.

2.2 REVIEW OF EXISTING LITERATURE

2.2.1 Review of Conventional Biomedical Waste Management Practices[1]

Farooq and Dharmendra presented a comprehensive review of traditional biomedical waste management practices followed in healthcare facilities. Their study explains that conventional systems mainly rely on manual segregation, collection, transportation, and disposal of waste. These methods vary across regions depending on available infrastructure and regulatory frameworks.

They emphasized that improper segregation of biomedical waste is a major issue in existing systems. When hazardous and non-hazardous waste are mixed, it increases environmental pollution and health risks. Poor disposal methods such as open dumping and uncontrolled incineration can lead to air, water, and soil contamination.

The authors suggested that traditional systems are no longer sufficient to handle the increasing volume of biomedical waste. They recommended the adoption of automated and smart technologies to reduce human involvement. Integrating monitoring systems can improve efficiency, safety, and overall waste management practices.

2.2.2 Challenges and Importance of Proper Waste Management[2]

Bansod and Deshmukh conducted a systematic review highlighting the importance of proper biomedical waste management. Their study explains that effective waste management is essential to prevent infections, protect healthcare workers, and maintain environmental safety.

They identified several key challenges in existing systems, such as lack of proper training among healthcare workers, inadequate infrastructure, and poor implementation of government regulations. Many healthcare facilities fail to follow standard guidelines, leading to unsafe handling and disposal of waste.

Their research also stresses the importance of awareness and education programs. Training healthcare staff and strictly enforcing policies can significantly improve waste management practices. They concluded that a combination of proper training, infrastructure development, and regulatory compliance is necessary for effective BMW management.

2.2.3 Application of Artificial Intelligence in Biomedical Waste Management[3]

Sarkar, Dey, and Malik explored the use of Artificial Intelligence in biomedical waste management systems. Their work focuses on improving waste segregation and monitoring through intelligent algorithms.

They proposed AI-based models that can automatically classify waste using image processing and machine learning techniques. These systems can identify different types of waste, such as metal and non-metal, with high accuracy. This reduces the dependency on manual segregation and minimizes human errors.

The study highlights that AI can significantly enhance the efficiency and reliability of waste management systems. By enabling real-time decision-making and automation, AI helps in achieving better safety standards and improved operational performance in healthcare facilities.

2.2.4 Role of Internet of Things in Smart Waste Monitoring Systems[4]

Mohamed, Khan, and Jagtap focused on the application of IoT in biomedical waste management. Their study demonstrates how IoT technology can be used to develop smart waste monitoring systems.

They introduced IoT-enabled smart bins equipped with sensors to track waste levels in real time. These systems send data to a central platform, allowing authorities to monitor bin status remotely. This helps in avoiding overflow and maintaining hygiene in healthcare environments.

Additionally, IoT improves waste collection efficiency by optimizing collection schedules. Instead of routine collection, bins are emptied only when needed. This reduces operational costs, saves time, and ensures a cleaner and safer waste management process.

2.2.5 Automated and Semi-Automated Waste Segregation Systems[5]

Somala Rama Kishore and his team proposed advanced methods for biomedical waste segregation using automation technologies. Their work focuses on reducing human involvement and improving safety during waste handling.

They developed systems that use sensors and mechanical components to automatically separate different types of waste. Semi-automated systems assist workers by simplifying the segregation process, while fully automated systems perform segregation with minimal human interaction.

The study emphasizes that automation can greatly reduce risks associated with manual handling of hazardous waste. It also improves accuracy, efficiency, and consistency in waste segregation. Their research supports the integration of modern technologies to achieve safer and more effective biomedical waste management systems.

2.3 COMPARISON OF EXISTING BIOMEDICAL WASTE MANAGEMENT SYSTEMS

Table 2.1 Comparison of Existing Biomedical Waste Management Systems

Author / Year	Technology Used	Key Contribution	Advantages	Limitations
Farooq & Dharmendra (2023)	Conventional + Smart Concepts	Review of BMW management practices	Highlights need for automation	No practical implementation
Bansod & Deshmukh (2023)	Traditional Methods	Focus on awareness and policy	Identifies real-world challenges	Lacks technological solutions
Sarkar, Dey & Malik (2023)	Artificial Intelligence	AI-based waste classification	High accuracy, reduced human effort	Expensive and complex
Mohamed, Khan & Jagtap (2023)	Internet of Things	IoT-based smart waste monitoring	Real-time tracking, efficient collection	Requires network infrastructure

2.4 RESEARCH GAP IDENTIFIED

- Lack of fully automated systems combining AI and IoT
- High dependency on manual labor
- Limited real-time monitoring
- Lack of cost-effective solutions for small healthcare centers
- Need for scalable and user-friendly systems

CHAPTER 3

EXISTING SYSTEM

Traditional waste disposal systems mostly rely on manual operation. In many places, people open the dustbin lid manually to dispose of waste. This process can spread germs and cause unhygienic conditions. Some automatic bins have been developed using simple sensors, but they do not provide waste segregation. Many systems only focus on opening the lid automatically without identifying the type of waste. As a result, different types of waste are mixed together. This makes recycling difficult and increases waste management costs. Existing systems also lack proper monitoring and automation. They may not detect metal waste or other specific materials. The absence of automatic segregation reduces efficiency in waste handling. Therefore, improved smart dustbin systems are needed to overcome these limitations.

3.1 MANUAL SEGREGATION OF BIOMEDICAL WASTE

In existing systems, biomedical waste segregation is primarily done manually by healthcare workers at the point of generation. Staff are responsible for identifying whether the waste is metal or non-metal and placing it in the appropriate bin. This process requires proper knowledge and awareness of waste categories.

However, manual segregation is highly dependent on human accuracy and consistency. Due to heavy workload, lack of training, or negligence, mistakes often occur. Improper segregation can lead to mixing of hazardous and non-hazardous waste, increasing health risks.

Moreover, manual handling exposes workers to dangerous materials such as sharps and infectious waste. This increases the risk of injuries and infections. Therefore, while manual segregation is simple, it is not always safe or reliable.

FIG 3.1 MANUAL SEGRETATION OF BIOMEDICAL WASTE

3.2 USE OF COLOUR-CODED BINS

Colour-coded bins are widely used in hospitals to simplify biomedical waste segregation. Each colour represents a specific type of waste, helping healthcare workers quickly identify where to dispose of items. This system supports organized and systematic waste management.

For example, metallic sharps are disposed of in puncture-proof containers, while non-metal waste like plastics and cotton is placed in designated bins. This visual method reduces confusion and improves efficiency in waste handling.

Despite its advantages, this system still relies on correct human usage. If workers fail to follow guidelines, waste may be disposed of incorrectly. Thus, colour-coding alone is not sufficient to ensure proper segregation.

FIG 3.2 COLOUR BINS

3.3 LACK OF AUTOMATED SEGREGATION SYSTEMS

Most existing biomedical waste management systems lack automation in the segregation process. The separation of metal and non-metal waste is usually not supported by machines or smart technologies. This limits the efficiency of the system.

Without automation, the process becomes slow and prone to errors. Human involvement at every stage increases the chances of incorrect disposal. This also affects the overall effectiveness of waste management in healthcare facilities.

Additionally, the absence of automation prevents real-time tracking and smart decision-making. Modern technologies like sensors, AI, and IoT are not widely implemented. This creates a gap between traditional methods and advanced solutions.

3.4 BIN LEVEL MONITORING SYSTEM

To prevent overflow of waste bins, some systems use bin level monitoring techniques. Sensors such as infrared (IR) or ultrasonic sensors are installed to detect the level of waste inside the bin. This helps in maintaining cleanliness.

When the waste reaches a certain level, the system provides an alert through LEDs or alarms. This ensures that the bins are emptied on time and reduces the chances of spillage. It also improves hygiene within healthcare facilities.

However, these systems are still basic and not fully integrated with advanced monitoring technologies. In many places, bin levels are still checked manually. This reduces the effectiveness of waste management systems.

3.5 LIMITATIONS OF EXISTING SYSTEM

- Heavy dependence on manual labour for waste segregation and monitoring
- High chances of human error and inefficiency
- Lack of advanced technology integration (IoT, AI, smart sensors)
- No real-time monitoring and tracking of waste
- Difficult to manage and control the overall process
- Risk of improper waste handling
- Causes environmental pollution
- Increases health risks for workers and the public
- Need for modernization using smart technologies

CHAPTER 4

PROPOSED SYSTEM

The proposed system introduces a smart dustbin capable of automatic waste detection and segregation. It uses an Arduino Uno microcontroller to control the entire system. An IR sensor detects the presence of waste near the dustbin. When waste is detected, the microcontroller activates a servo motor to open the lid automatically. After the waste enters the bin, an inductive proximity sensor checks whether the material is metallic. If metal is detected, the system directs the waste to the metal compartment. Otherwise, the waste goes to the general waste bin. The servo motor helps control the lid and internal segregation mechanism. This system improves hygiene by reducing physical contact with waste. It also helps in proper waste classification. The proposed design is simple, cost-effective, and easy to implement. It can be used in hospitals, offices, public areas, and residential buildings.

4.1 ARCHICTECTURE OF PROPOSED SYSTEM

The proposed system is a smart dustbin designed to improve hygiene and waste management through automation. The system uses sensors and a microcontroller to detect waste and perform basic waste segregation. The main objective of the system is to reduce human contact with waste and make the disposal process more efficient.

In this system, an IR sensor is used to detect the presence of waste near the dustbin. When a person brings waste close to the bin, the sensor sends a signal to the Arduino microcontroller. The microcontroller processes the signal and activates the servo motor to open the lid automatically. This allows the user to dispose of waste without touching the dustbin.

After the waste enters the dustbin, an inductive proximity sensor is used to detect whether the waste material is metallic or non-metallic. If metal is detected, the microcontroller directs the waste into the metal compartment. If the waste is non- metallic, it is directed into the general waste compartment.

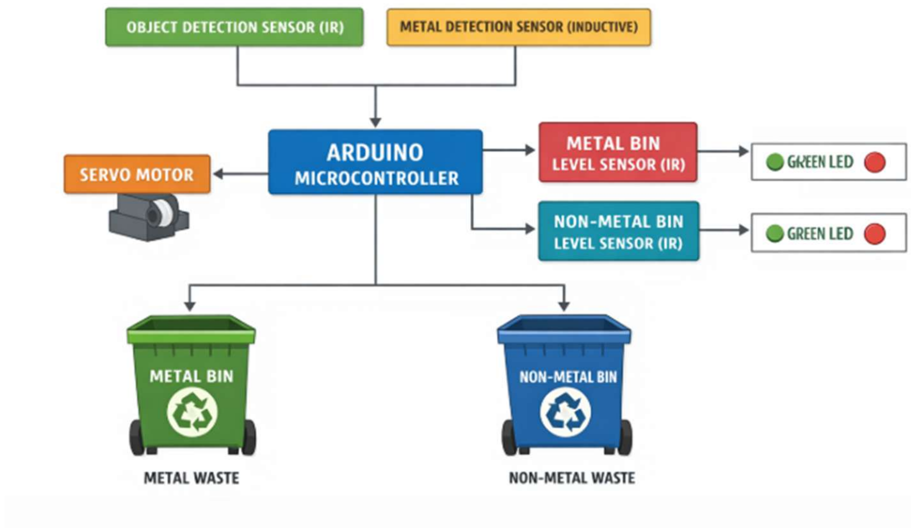

FIG 4.1 ARCHICTECTURE OF PROPOSED SYSTEM

4.2 WORKING PRINCIPLE OF PROPOSED SYSTEM

This diagram illustrates an Arduino-based automatic waste segregation system that efficiently separates metal and non-metal waste. The system uses an IR object detection sensor to detect when waste is placed, along with a metal detection (inductive) sensor to identify whether the material is metallic or non-metallic. These sensors send signals to the Arduino microcontroller, which acts as the central processing unit.

The Arduino processes the input data and controls a servo motor to sort the waste accordingly. If the object is detected as metal, the servo directs it into the metal bin; if it is non-metal, it is directed into the non-metal bin. This automated sorting mechanism ensures accurate and quick separation without human intervention.

Additionally, each bin is equipped with an IR level sensor to monitor how full it is. The system uses LED indicators to display the bin status, where a green LED shows that the bin has space and a red LED indicates that the bin is full. This setup improves waste management efficiency, reduces manual effort, and supports better recycling practices.

4.3 PROPOSED SYSTEM FLOW CHART

FIG 4.2 PROPOSED SYSTEM FLOW CHART

The system workflow describes the sequence of operations in the smart dustbin system. First, the IR sensor continuously monitors the area near the dustbin. When a user brings waste close to the bin, the sensor detects the object and sends a signal to the Arduino.

The Arduino processes the signal and activates the servo motor to open the lid automatically. The user can then dispose of the waste into the dustbin.

After the waste enters the bin, the inductive proximity sensor checks whether the waste material is metallic. If metal is detected, the Arduino directs the waste to the metal compartment. Otherwise, it goes to the general waste compartment.

Finally, the servo motor closes the lid after a short delay. This workflow ensures smooth and automatic waste disposal and segregation.

4.4 FEATURES AND BENEFITS

The smart dustbin system has several important features that improve waste management. One of the main features is automatic lid operation using sensors. This reduces human contact with waste and improves hygiene.

The system is cost-effective and easy to implement using simple electronic components. It also consumes low power and requires minimal maintenance. The smart dustbin can be used in hospitals, offices, and public places to maintain cleanliness. It also supports recycling efforts by separating waste materials.

4.5 ADVANTAGES OF PROPOSED SYSTEM

- Fully automatic system reduces manual effort
- Touchless operation improves hygiene
- Prevents spread of germs and bacteria
- Automatic lid mechanism for easy waste disposal
- Performs basic waste segregation (metal and general waste)
- Improves recycling efficiency
- Simple design and easy to use

CHAPTER 5
TOOLS REQUIRED

The smart dustbin project requires both hardware and software tools for implementation. The main hardware component is the Arduino Uno microcontroller. Sensors such as IR sensors and inductive proximity sensors are used to detect waste and identify materials. A servo motor is used to control the lid movement and waste direction. Jumper wires and a breadboard are used for circuit connections. A power supply is required to operate the system. The Arduino IDE software is used to write and upload the program to the microcontroller. The development environment allows users to compile and debug the code easily. Hardware assembly involves connecting sensors and motors to the Arduino board. Proper configuration ensures that all components work together efficiently. These tools help build and test the smart dustbin system.

5.1 HARDWARE TOOLS

The development of the smart dustbin system requires both hardware and software tools. The primary hardware component used in the project is the Arduino Uno microcontroller, which controls the entire operation of the system. Sensors such as the IR sensor and inductive proximity sensor are used to detect waste and identify metal materials. A servo motor is used to control the opening and closing of the dustbin lid and the segregation mechanism.

The software tool used for programming is the Arduino IDE. This software allows users to write, compile, and upload code to the Arduino board. The Arduino programming language is based on C and C++, which makes it easy for beginners to learn. The IDE also provides a serial monitor to check the output and debug the program.

These tools and technologies work together to create an automated and intelligent waste management system. The use of simple electronic components makes the system affordable and easy to implement.

5.1.1 Arduino Uno

The Arduino Uno is the main controller used in the smart dustbin system. It is based on the ATmega328P microcontroller and is widely used in embedded system projects. The Arduino board receives input signals from sensors and processes them according to the programmed instructions.

The Arduino Uno has several digital input and output pins that allow easy connection of sensors and motors. It also supports communication with other devices through serial communication. The board can be programmed using the Arduino IDE software.

In this project, the Arduino reads the signals from the IR sensor and inductive proximity sensor. Based on the sensor outputs, it sends commands to the servo motor. The microcontroller continuously monitors the sensors and controls the system automatically. Because of its simplicity, low cost, and reliability, the Arduino Uno is suitable for developing smart automation projects.

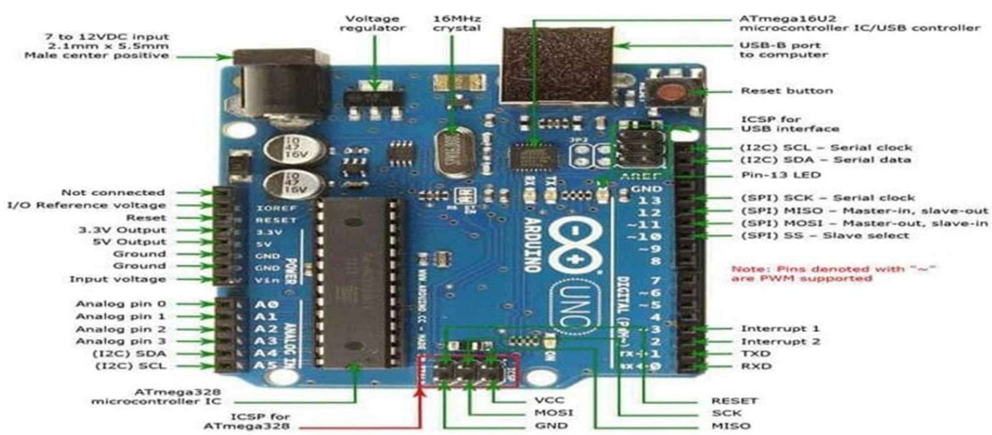

FIG 5.1 ARDUINO UNO

5.1.2 IR Sensor

The IR Sensor Module is used to detect the presence of objects near the dustbin. It works using infrared light. The sensor emits infrared radiation and detects the reflected signal when an object comes close to it.

When a person brings waste near the dustbin, the IR sensor detects the object and sends a signal to the Arduino microcontroller. The microcontroller then activates the servo motor to open the lid of the dustbin.

In the smart dustbin system, the IR sensor helps create a touch-free waste disposal mechanism. This improves hygiene and makes the dustbin easier to use in public places.

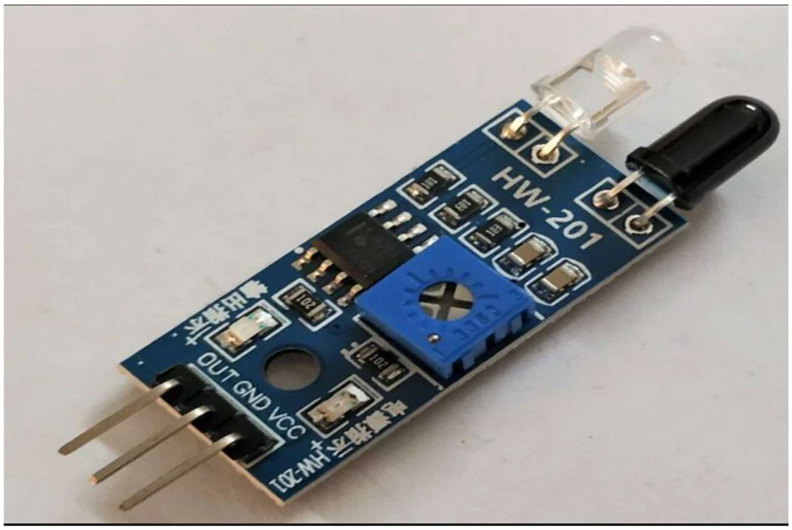

FIG 5.2 IR SENSOR

5.1.3 Inductive Proximity Sensor

The Inductive Proximity Sensor is used to detect metallic objects. It works based on electromagnetic induction principles. When a metal object is placed near the sensor, it creates a change in the electromagnetic field, which is detected by the sensor.

The sensor sends a signal to the Arduino microcontroller when metal is detected. The microcontroller then processes this signal and directs the waste into the metal compartment.

Inductive proximity sensors are commonly used in industrial automation systems for metal detection. They are reliable and can detect metal objects without direct contact.

In the smart dustbin project, the inductive sensor plays an important role in waste segregation. It helps separate metal waste from other types of waste.

FIG 5.3 INDUCTIVE PROXIMITY SENSOR

5.1.4 Servo Motor

The Servo Motor SG90 is used to control the movement of the dustbin lid and internal waste segregation mechanism. A servo motor can rotate to a specific angle based on the signal received from the Arduino microcontroller.

In this project, the servo motor opens the lid when waste is detected by the IR sensor.After the waste is disposed of, the lid closes automatically after a short delay.The servo motor can also rotate to direct waste into different compartments based on sensor detection. This helps achieve automatic waste segregation.

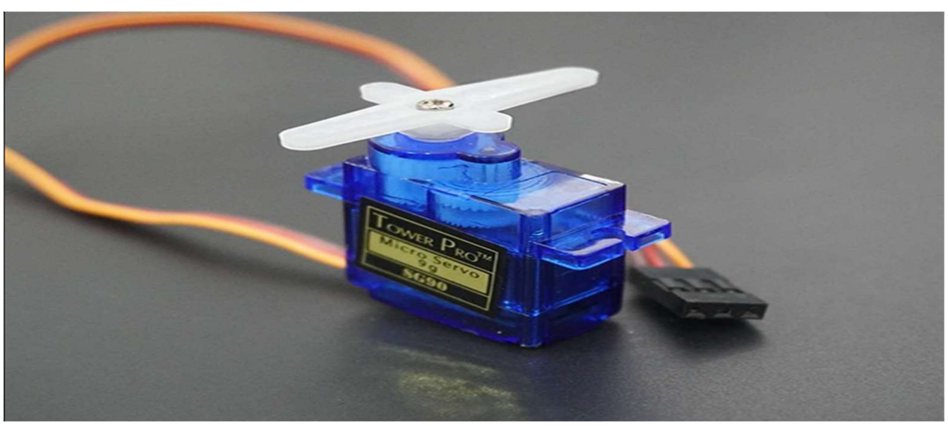

FIG 5.4 SERVO MOTOR

5.1.5 Power Supply

A power supply is required to operate the smart dustbin system. The power supply provides electrical energy to the Arduino board, sensors, and servo motor. In most cases, a 5V power supply is used to power the Arduino Uno.

The sensors and servo motor also require stable voltage for proper operation. Power can be provided using a battery, adapter, or USB connection.

A reliable power supply ensures that all components function correctly without interruption. Proper voltage regulation is important to prevent damage to electronic components.

In this project, the power supply enables the sensors and microcontroller to perform detection and control tasks continuously.

5.1.6 LED

LED is used as a visual indicator to show the type of waste detected. When the system identifies metal waste, the metal LED glows, and when non-metal waste is detected, the non- metal LED turns ON.

This helps the user easily understand the classification process. The LED can also indicate system status, such as whether the system is active or if a bin is full, making the operation clear and user-friendly.

FIG 5.5 LED

5.1.7 Buzzer

The buzzer is used to provide audio alerts for different system conditions. First, it produces a sound when waste is detected, indicating that the system has started processing. Second, it gives a confirmation sound after successful segregation of metal or non-metal waste. Third, it acts as a warning alert when the bin becomes full or if there is any issue in the system.

These sound indications make the system more interactive and help the user stay informed without constantly watching it.

FIG 5.6 Buzzer

5.2 SOFTWARE TOOLS

The development of the smart dustbin system requires various software tools for programming, simulation, and testing. The primary tool used is Arduino IDE, which is used to write, compile, and upload code to the Arduino microcontroller. It provides a simple interface and supports multiple libraries for sensors, motors, and LCDs. Embedded C is the programming language used within the Arduino IDE to control hardware components. For circuit design and simulation, tools like Proteus or Tinkercad can be used to test the system virtually before hardware implementation. These tools help in identifying errors and improving design efficiency. Serial Monitor in Arduino IDE is used for debugging and displaying real-time sensor data.

5.2.1 Arduino IDE

Arduino IDE (Integrated Development Environment) is a software application used to write, compile, and upload programs to Arduino boards. It provides a simple and user-friendly interface, making it suitable for both beginners and professionals working on embedded systems and IoT projects.

The IDE includes an editor window where users can write code, a toolbar with options to verify and upload programs, and a serial monitor to display real-time data for debugging. The working process involves writing the code (called a sketch), compiling it to check for errors, selecting the appropriate board and port, and uploading it to the Arduino using a USB connection.

Once uploaded, the Arduino executes the program. Overall, Arduino IDE is widely used due to its simplicity, open-source nature, and strong community support, making it ideal for applications like IoT, robotics, home automation, and smart waste management systems.

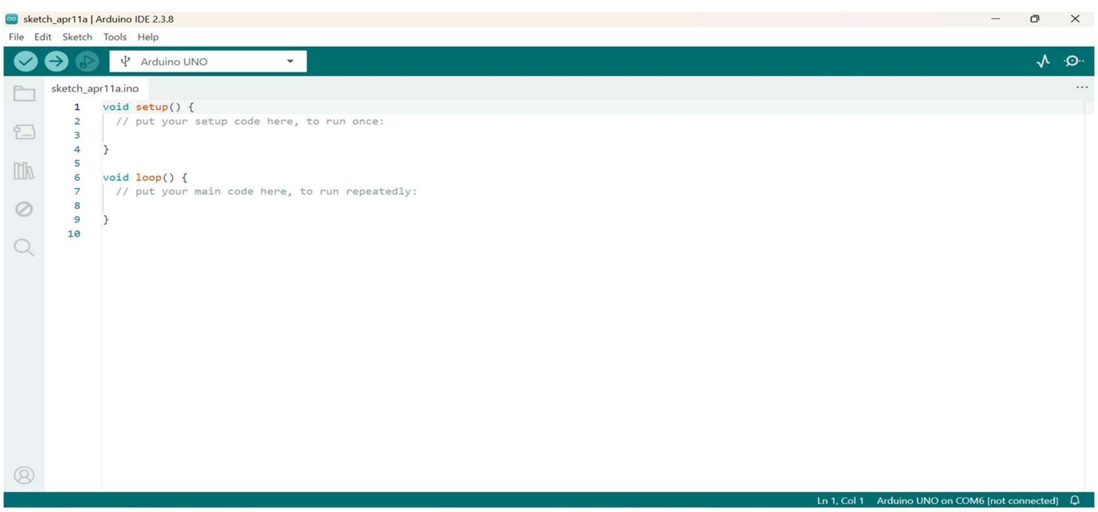

FIG 5.7 ARDUINO IDE

CHAPTER 6
RESULTS AND DISCUSSION

The results of the smart dustbin system demonstrate that the sensors and microcontrollers operate effectively and reliably. The IR sensor successfully detects the presence of waste when it is placed near the bin and sends a signal to the Arduino microcontroller.

Based on this input, the servo motor automatically opens the lid, enhancing user convenience and maintaining hygiene. After the waste is deposited, the inductive proximity sensor identifies whether the object is metallic or non-metallic. If a metallic object is detected, the system directs it to the metal compartment; otherwise, it is directed to the general waste bin.

The Arduino processes all sensor inputs accurately and controls the servo motor accordingly. The system performs waste detection and segregation efficiently under real-time conditions.

Overall, the performance of the system is consistent and accurate, demonstrating that the proposed design can significantly improve modern waste management practices.

6.1 IMPLEMENTATION OF PROPOSED SYSTEM

The results obtained from the smart dustbin system show that the sensors function correctly. The IR sensor successfully detects the presence of waste near the dustbin. When an object is placed close to the sensor, it sends a signal to the Arduino microcontroller.

The inductive proximity sensor also performs accurately in detecting metallic objects. When a metal object is placed inside the dustbin, the sensor identifies it and sends a signal to the microcontroller.

The Arduino processes these signals and controls the servo motor accordingly. The detection results confirm that the sensors are capable of performing waste detection and material identification effectively.

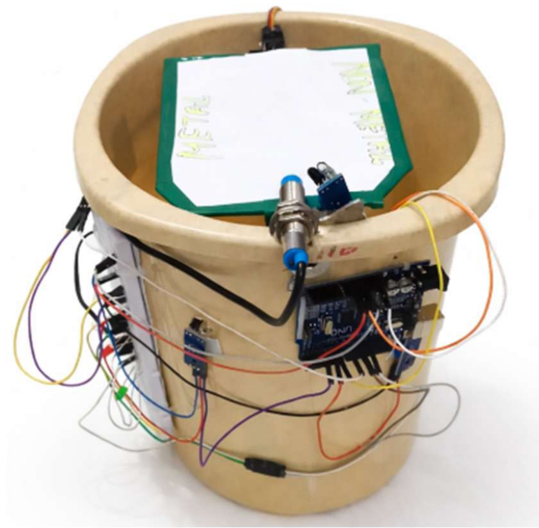

FIG 6.1 IMPLEMENTATION OF PROPOSED SYSTEM

6.2 WORKING OF PROPOSED SYSTEM

The servo motor plays an important role in the operation of the smart dustbin. When the IR sensor detects waste, the Arduino sends a signal to the servo motor to open the lid. The lid opens smoothly and allows the waste to be disposed of.

After the waste is deposited, the lid closes automatically after a short delay. The servo motor also helps direct the waste into the appropriate compartment based on the sensor detection.The motor response is quick and accurate, which ensures smooth operation of the system. The successful actuation of the servo motor confirms that the automated lid mechanism works properly.

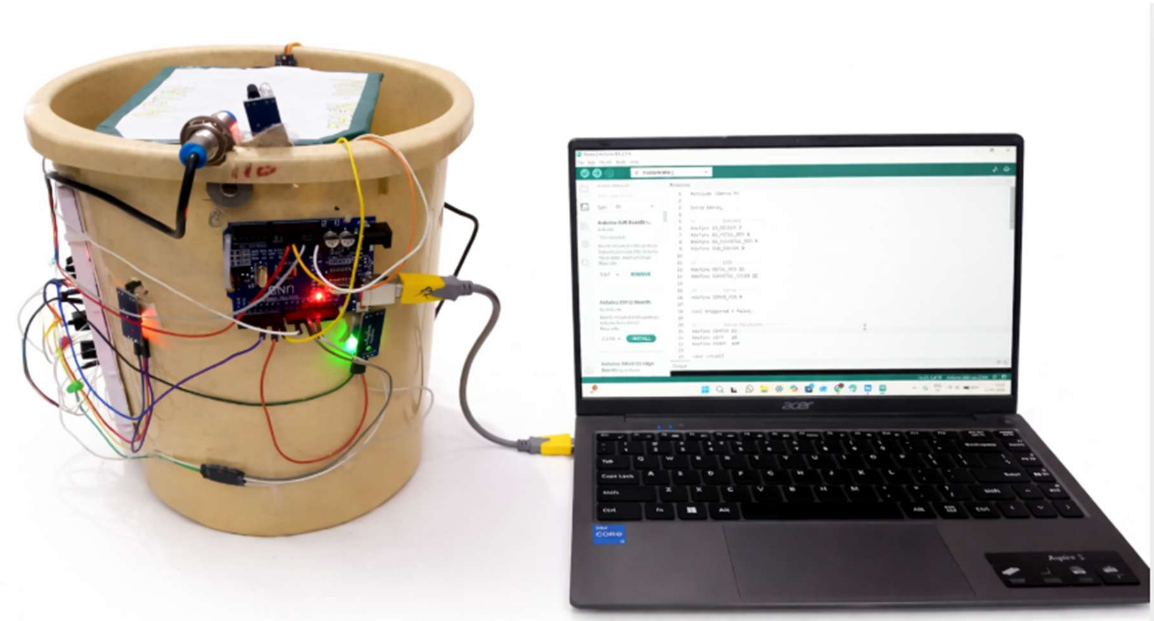

FIG 6.2 WORKING OF PROPOSED SYSTEM

6.3 ANALYSIS OF EXISTING VS PROPOSED SYSTEM FUNCTIONALITIES

Table 6.1 Analysis of Existing vs Proposed System

S.No	Parameter	Existing System (Traditional Dustbin)	Proposed System (Smart Dustbin)
1	Power Supply	No power required (manual operation).	Uses Arduino power supply (USB/battery 5V– 12V).
2	Functionality	Only collects waste manually.	Automatic lid opening and waste segregation system.
3	Input Devices	No sensors used.	IR sensor for object detection
4	Output Devices	No output indication.	Servo motor for lid control and segregation mechanism.
5	Communication	No communication system.	Works standalone embedded system.
6	Security Features	No safety or hygiene features.	Touch-free improves hygiene and safety.
7	Data Storage	No data storage.	No storage required (real- operation).
8	Target Application	Household and Public waste collection.	Smart cities, hospitals asnd automated waste management systems.
9	Scalability	Cannot be upgrade	Can be extended with IoT and additional sensors.
10	Energy Efficiency	No energy usage.	Low power consumption Using efficient components.

11	Accessibility	Requires manual handling.	User-friendly and touchless operation.
12	Segregation Method	No waste segregation.	Automatic segregation (metal vs non-metal).
13	Ease of Use	Simple but unhygienic.	Easy to use and hygienic.

The existing traditional dustbin system operates without any power supply and relies completely on manual handling for waste collection, making it simple but unhygienic. It does not use any input devices such as sensors and lacks output components, meaning there is no automation or indication system. Additionally, there is no communication mechanism, data storage, or safety features, which increases the risk of improper waste handling and poor hygiene. Waste segregation is not performed, and the system cannot be upgraded or scaled for advanced applications.

It is mainly used for basic household and public waste collection but requires continuous human effort. In contrast, the proposed smart dustbin system uses a power supply (5V–12V via Arduino) and incorporates automation to improve efficiency and hygiene. It uses an IR sensor to detect waste and a servo motor to control the lid and perform basic waste segregation between metal and non-metal.

The system operates as a standalone embedded setup and provides touch-free functionality, enhancing safety and cleanliness. Although it does not require data storage, it performs real-time operations effectively. The system is user-friendly, energy-efficient, and suitable for applications such as smart cities, hospitals, and automated waste management systems. Furthermore, it is scalable and can be upgraded with IoT and additional sensors, making it a modern and efficient alternative to traditional dustbins.

CHAPTER 7

CONCLUSION AND FUTUREWORK

7.1 CONCLUSION

The smart dustbin system developed in this project provides an efficient solution for automated waste management. The system uses sensors, a microcontroller, and a servo motor to detect waste and perform basic segregation.The IR sensor allows the dustbin lid to open automatically when waste is detected. The inductive proximity sensor helps identify metallic waste and separate it from other materials. The Arduino microcontroller controls all system operations and ensures smooth functioning.The system reduces manual effort and improves hygiene by providing a touch- free waste disposal mechanism. It also promotes recycling by separating metal waste. Overall, the smart dustbin is a simple and cost-effective solution for improving waste management.

7.2 FUTURE WORK

The smart dustbin system can be further improved by adding advanced technologies. In the future, additional sensors can be used to detect different types of waste such as plastic, paper, and organic materials.

The system can also be integrated with IoT technology to monitor the dustbin status remotely. This would allow authorities to track waste levels and schedule collection efficiently.

Solar power can also be used to make the system more energy-efficient and environmentally friendly. Adding wireless communication modules could further enhance the system's functionality.

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APPENDIX

Arduino IDE

Main code:

```
#include <Servo.h>

Servo servo;

// -------- Sensors --------
#define IR_OBJECT 7
#define IR_METAL_BIN 6
#define IR_NONMETAL_BIN 4
#define IND_SENSOR 8

// -------- LEDs --------
#define METAL_GREEN 10
#define METAL_RED 11

#define NONMETAL_GREEN 12
#define NONMETAL_RED 13

// -------- Servo --------
#define SERVO_PIN 9

bool triggered = false;

// -------- Servo Positions (ADJUSTED) --------
#define CENTER 61 // shifted LEFT (was 60)
#define LEFT   10
#define RIGHT  120
```

```
void setup()
{
 pinMode(IR_OBJECT, INPUT);
 pinMode(IR_METAL_BIN, INPUT);
 pinMode(IR_NONMETAL_BIN, INPUT);
 pinMode(IND_SENSOR, INPUT);

 pinMode(METAL_GREEN, OUTPUT);
 pinMode(METAL_RED, OUTPUT);

 pinMode(NONMETAL_GREEN, OUTPUT);
 pinMode(NONMETAL_RED, OUTPUT);

 servo.attach(SERVO_PIN);
 servo.write(CENTER); // corrected center

 Serial.begin(9600);
}

void loop()
{
 int objectDetect = digitalRead(IR_OBJECT);

 // -------- OBJECT DETECTED --------
 if(objectDetect == LOW && triggered == false)
 {
  triggered = true;

  delay(300);

  int metal = digitalRead(IND_SENSOR);

  // -------- METAL --------
  if(metal == LOW)
  {
   Serial.println("Metal Detected");
```

```
    servo.write(LEFT);
    delay(700);
  }

  // -------- NON METAL --------
  else
  {
    Serial.println("Non Metal Detected");

    servo.write(RIGHT);
    delay(700);
  }

  // Back to corrected center
  servo.write(CENTER);
  delay(600);
}

// -------- RESET --------
if(objectDetect == HIGH)
{
  triggered = false;
}

// -------- BIN LEVEL CHECK --------
int metalBin = digitalRead(IR_METAL_BIN);
int nonMetalBin = digitalRead(IR_NONMETAL_BIN);

// Metal bin
if(metalBin == LOW)
{
  digitalWrite(METAL_RED, HIGH);
  digitalWrite(METAL_GREEN, LOW);
}
else
{
  digitalWrite(METAL_RED, LOW);
  digitalWrite(METAL_GREEN, HIGH);
```

```
  }

  // Non-metal bin
  if(nonMetalBin == LOW)
  {
    digitalWrite(NONMETAL_RED, HIGH);
    digitalWrite(NONMETAL_GREEN, LOW);
  }
  else
  {
    digitalWrite(NONMETAL_RED, LOW);
    digitalWrite(NONMETAL_GREEN, HIGH);
  }

  delay(50);
}
```

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Dr. Getsy S Sara
Event Co-ordinator

Dr. U.V.Arivazhagu
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